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Simulation Model of a Quality Control Laboratory in Pharmaceutical Industry ⋆

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Abstract: Laboratories are critical components in drug manufacturing, and inefficiencies in laboratory management can have a major impact on the overall supply chain service level. The aim of this paper is to build a Discrete Event Simulation model of a Quality Control laboratory. To achieve this objective, a generic framework for information treatment and organization was built. In particular, information coming from different databases was organized into a single one that was used as input to a discrete event simulation model. The proposed model represents in detail the work flow of a quality control laboratory and it is intended as a support tool for planning, scheduling and decision making. The model was validated using real data, and it proved to be effective in the estimation of performance parameters such as, system throughput, equipment usage rate, system responsiveness and tasks processing times. Furthermore, the simulation model was tested with an alternative scheduling policy to evaluate how modifications on the system may improve its performance.

Keywords: Computer Simulation, Discrete Event Systems, Modeling, Quality Control, Scheduling, Pharmaceutical Industry

1. INTRODUCTION

In the last decades, drug manufacturing evolved driven by external economic forces, patents expiration and increased competition. In order to maintain their competitive advantage, pharmaceutical companies organized themselves into complex organizations (supply chain and contract manufacturing networks), and started to be more concerned on achieving operational excellence through the optimization of the processes involved in drug manufacturing: chemical processes modeling, supply of raw materials, logistic operations, quality control, etc.

Pharmaceutical companies operate in one of the most competitive and regulated markets, and compliance with Good Manufacturing Practices (GMP) and Good Laboratory Practices (GLP) is mandatory for marketing any drug. Those regulations were introduced to ensure that every pharmaceutical product meets safety and quality requirements in a systematic fashion.

During its development life cycle, a drug must be constantly monitored with laboratory tests to assess its quality. In this scenario, quality control laboratories play an important role in the drug manufacturing process. Laboratory management is a complex task which involves resources (personnel and equipment) planning and scheduling, analysis prioritization, results evaluation and documentation.

Quality control laboratories are critical components in drug manufacturing and inefficiencies in quality control may delay obtaining results, affect negatively their quality and can have a major impact on the overall supply chain service level. The situation can be magnified in the case of a contract manufacturing organization, that produces goods under the brand of its clients and therefore has to deal with a large number of projects and materials.

Given the high mix of products and tests it is important to develop effective strategies for laboratory management. Laboratory information management systems (LIMS) used in pharmaceutical industry often lack of features essential (i.e. information on processing times, work flow) for planning, scheduling and stock management.

As pointed out by Juran and Godfrey (1999), quality control can be seen as a recursive process composed by three main tasks: *quality planning, control and improvement*. Thus, quality control is not only responsible for the execution of quality tests but also for the development and the improvement of methods to be used in the control process. Dedicated optimization methods for planning and scheduling, in addition with a generic framework for infor-

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mation management can improve laboratory performances and resource usage.

1.1 Related Work

In the pharmaceutical industry, laboratory management is often done based on supervisors' experience, rather than with validated data driven models. Literature is short of examples of generic optimization frameworks applicable to pharmaceutical quality control laboratory management.

Klaessens et al. (1988) developed a simulation model that consists of classes of objects (sample, sample generator, planner, analyst and instruments) and applies concepts of queuing theory to describe a laboratory organization. The authors proposed two planning strategies: centralized (the workload of a day is planned by the expert planner) and decentralized (the analysts determine which samples are analyzed).

Schäfer (2004) proposes a set of terms and definitions that can be used to model all components interacting on the workbench (i.e. samples, instruments, sensors, results, information systems) and provides a description of a schematic scheduling process applicable in quality control laboratories. His approach comprises the following steps: *process description*, *information treatment*, *work-plan generation*, *scheduling generation*, *execution*, *instrumental control and data storage*.

Maslaton (2012) presents a two steps process towards the optimization of a quality control laboratory. The first step is related with resource planning, while the latter refers to the scheduling of the laboratory. The author proposes to organize information on products and raw materials into groups, based on products similarities, define types and number of tests for each group and estimate analysis processing time for each test. Once the correct amount of resources has been determined the second step of the optimization process is the scheduling of the analyses.

To assign the best available resource to a sample/test, classical optimization algorithms can be used effectively. Boyd and Savory (2001) applied a genetic algorithm for laboratory personnel scheduling. Their program maximizes the value of a fitness function that measures how well a given work shift scheduling of analysts and their skills matches a set of work tasks. Similarly, Dudnikov et al. (2012) developed a laboratory analysis planning system to schedule analyses on the available resources. Their algorithm considers two types of analyses (*periodic* and *unplanned*), and creates a 24 hours scheduling considering only periodic samples that is rearranged in case of unplanned samples.

1.2 Objectives

The objective of this work is to build a generic framework for modeling and simulating a quality control laboratory of a pharmaceutical company. In particular, the modeling stage involves the study of the system and the design of an integrated database that contains all the relevant information that can be used for planning and scheduling. Moreover, it uses concepts of Discrete Event System (DES) and Petri net graphs theory for modeling purposes. See Cassandras and Lafortune (2009) and Law (2015).

A model of the real system is simulated using the academic license of Simio simulation software (version 8.132). The developed model will serve as basis for planning, scheduling and decision making in quality control laboratories.

2. QUALITY CONTROL LABORATORY OVERVIEW

Hovione Farmaciência S.A. is an international company operating in the contract manufacturing business sector. Quality control represents an important part of the drug manufacturing process. With thousands of samples to be analyzed every year, laboratories management can be difficult and inefficiencies at quality control level may affect negatively the overall supply chain service level.

Among several quality control laboratories, this study focuses on a particular one, that is located in the Portuguese facility of Loures (Lisbon). It is furnished with 70 equipment of 6 different types and employs 20 analysts working on 3 work shifts. The six classes of equipment are related with particular analysis techniques: *High Performance Liquid Chromatography* (HPLC), *Gas Chromatography* (GC), *Particle Size Analysis* (PSA), *Karl Fischer Titration*, *Differential Scanning Calorimetry* (DSC) and *X-Ray Powder Diffraction* (XRPD).

Typical tasks in pharmaceutical quality control laboratories are analytical methods development and validation, tests on final products, raw materials, in-process control and stability analyses. Analytical methods must be designed, developed and validated with the goal to rapidly test a specific compound.

2.1 Laboratory Staff and Shifts

The quality control lab considered in this study is open 24 hours a day, 7 days a week. A reason for this is the need to have always analysts available to support the manufacturing process with in-process control tests. Other types of tests (i.e. stability, development, validation) are performed 5 days a week, from Monday to Friday. A total of twenty analysts have been considered: six analysts work on 12 hours rotating shifts, four analysts on two different 8 hours shifts, and ten analysts work 5 days a week on a fix 8 hours shift. See table 1.

Table 1. Analysts Work Shifts.

Work Shift	Days a week	Hours shifts	Analysts on Shift
Shift 1	7	8:00-20:00 20:00-8:00	2
Shift 2	5	8:00-17:00 17:00-24:00	2
Shift 3	5	8:00-17:00	10

2.2 Analysis Work flow

Every day a high number of samples enters the laboratory to be analyzed. According to the analytical technique needed to analyze a particular sample, it flows to the appropriate group of machines and it is scheduled to one of the available machines. Independently from the analytical technique used, an analysis starts with the preparation of solutions and materials, and with the setup of the equipment. Next two steps involve the preparation of the sample and the verification of the equipment (*system suitability*). This step is needed to check if the equipment is suitable to run a specific test. Finally, the sample is analyzed and analysis data are processed by an analyst (see figure 1).

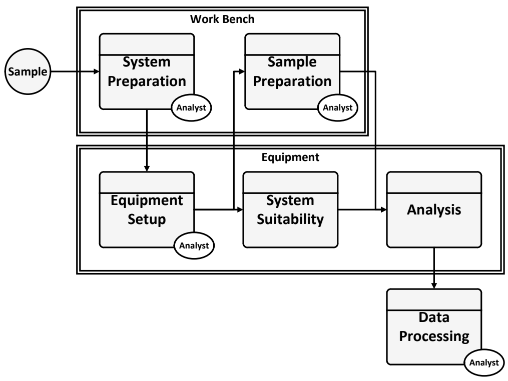

Fig. 1. Main tasks of a quality control test.

3. QC LABORATORY MODEL DESIGN

The goal of this study is to evaluate the performances of the quality control laboratory and to propose useful tools for planning and scheduling. The proposed methodology considers analysts and equipment as processing units (or resources). Jobs and activities are defined by the sample and by the particular analytical method needed to test it. Thus, samples can be considered as the input of the system. In the simulation software paradigm a sample can be defined as an entity that flows through the system and need to perform a number of activities at particular working stations.

3.1 Data Processing

In the section above, the main components of the system have been identified: analysis work flow, entities (samples) and processing units (analysts and equipment). The next step is to estimate *entities arrival rate*, *process logics* and *processing times*. With this purpose, a study of the system and of the available data was performed. Information was distributed on different databases, and in some cases it was stored in document repositories in unstructured form. Thus, it was necessary to extract and organize the relevant information coming from these data sources.

Four different data sources have been considered. The first one is the *Laboratory Information Management System* (LIMS) that contains information on all the analyses performed, such as number of samples processed, analytical methods used, equipment used, release date and completed date. However, LIMS has no information about *processing times* and *process logics*. Another source of information is the *document mangament system* (DMS) that contains reports with the description of all the analytical methods. Given the unstructured form of the data, an *information extraction algorithm* (IE) was implemented to analyze text and extract information on: analytical method name, number of samples to be prepared, system suitability and analysis run time. Furthermore, a *time study* was designed to gather information on tasks processing times and to divide equipment "hands-on" and "hands-off" tasks. With this purpose, it was asked to analysts to collect data during a period of one month using a form model. All this information was organized into a unique database, whose UML representation is shown in figure 2.

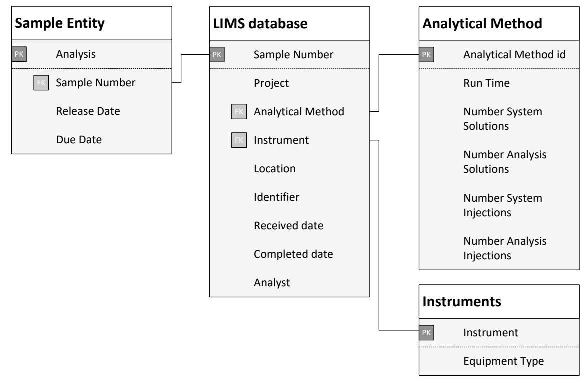

Fig. 2. UML representation of the designed database.

3.2 Input Analysis

The database structure presented allows to easily access to relevant information that can be used for modeling. In a simulation model it is important to define probability distributions for input in order to simulate the randomness existing in a real system. With this purpose, LIMS database offered good quality information to estimate samples arrival rate, and number of analyses for each analytical technique. LIMS provides an *arrival table* where each entry represents the couple analysis/sample and can be considered as input to the system. In table 2 it is shown the percentage of analyses per analytical technique.

Table 2. Percentage of analyses per technique.

Technique	*HPLC*	*GC*	*PSA*	*KF*	*DSC*	*XRPD*
Percentage	40.7%	25.7%	18.8%	7.1%	4.0%	3.7%

Many of the processing times in the model are stochastic in nature, while others are deterministic. Equipment run time and number of preparations have been retrieved from analytical methods reports, and thus are assumed to be deterministic. Processing times for tasks performed by the analysts (i.e. equipment setup, sample preparation, data processing) were estimated fitting probability distributions to data gathered with the *time study*. Processing times distribution used in simulation are summarized in table 3.

Table 3. Processing times distributions. Time unit is minutes.

Equip.	**Sample Prep.**	**Equip. Setup**	**Run Time**	**Data Process**
HPLC	$T(5, 40, 100)$	$T(25, 50, 65)$	*Det.*	$T(2, 5, 10)$
GC	$T(5, 10, 35)$	$T(2, 5, 30)$	*Det.*	$T(10, 20, 45)$
PSA	$T(5, 7.5, 21)$	$U(1, 32)$	$T(5, 10, 30)$	$T(2, 6, 10)$
KF	$T(1, 6, 25)$	$T(4, 8, 16)$	$T(2, 5, 56)$	$T(10, 20, 30)$
DSC	$T(5, 10, 30)$	$T(1, 3, 12)$	$U(12, 180)$	$T(4, 9, 16)$
XRPD	$T(5, 10, 20)$	$T(1, 5, 20)$	$T(11, 25, 203)$	$T(2, 7, 30)$

Given the empirical nature of the collected data, three types of probability distribution are used: deterministic

($Det.$), triangular pdf $T(min, mode, max)$ and uniform pdf $U(min, max)$.

3.3 Discrete Event Simulation Model

The simulation model was developed under the following assumptions:

(1) *The couple sample/analysis has been considered as a single entity.* Every sample entity is processed using only one machine;
(2) *The system suitability has a validity of 24 hours.* The system suitability does not need to be re-executed within 24 hours, unless a new sample using a different analytical method is scheduled to the same machine;
(3) *Analysts can perform every type of work regardless the work shift they are allocated to.* In the real system every work shift is allocated only to few operation types (i.e. in-process control, stability, development).

The quality control simulation model was built using Simio simulation software, and it is composed by four types of objects: samples, machines, analysts and work location nodes.

Samples have been modeled as input entities that travel through the system to perform certain tasks. Each sample has properties that define the type of equipment needed to process it, and additional information about run time and number of preparations.

A machine can be seen as a series of processes needed to perform an analysis (system preparation, system suitability, sample preparation, analysis and data processing). These tasks have to be performed at different physical location (bench, machine or computer location). The spatial component has been added to those processes as an attribute (work location node) that specifies the physical place in the laboratory where an analyst has to go to perform the requested task.

An analyst has been modeled as a secondary resource of the system, and his/her presence is necessary to perform specific tasks. As referred, analysts work in shifts and their availability in the laboratory varies according to the work shifts defined in table 1.

Generic Equipment Model A generic machine model has been implemented to simulate the analysis process. This model is valid for every analytical technique regardless of the equipment used. Thus, the analysis process was modeled using the concepts of discrete event systems (DES) theory and it was represented using timed Petri nets formalism.

A timed Petri net is a bipartite graph with two types of nodes (*places* and *transitions*) and a set of weighted arcs that links places to transitions and transitions to places. Places contains information about the state of a Petri net. This state is expressed by the *marking* function that assigns a number of tokens to each place of the Petri net. A transition is usually associated with an event that can be asynchronous or time dependent. An event can "happen" only if the relative transition is enabled. See Cassandras and Lafortune (2009).

The timed Petri net used to design the generic equipment model (figure 3) is characterized by a set of places P, that represents the five analysis tasks described above, more some auxiliary places needed to represent queues and to avoid impossible system dynamics (i.e. perform the system suitability and run the analysis at same time). When a sample enters the system it waits in a queue and it is scheduled according to a policy that tends to minimize the maximum lateness (calculated as the difference between the estimated completion time and the due date). This scheduling policy is known as *maximum lateness first*). See Pinedo (2015).

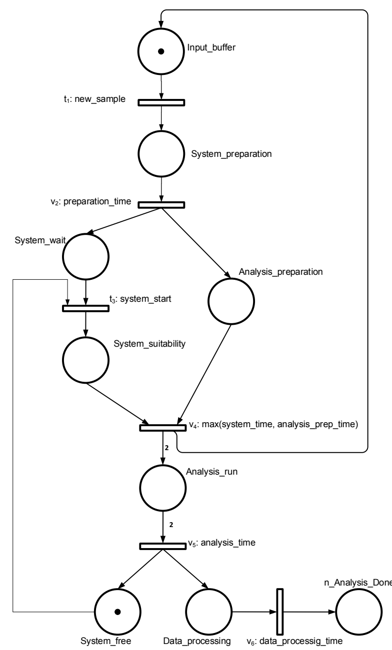

Fig. 3. Discrete Event System model of a generic equipment using timed Petri net.

Quality Control laboratory Model The described model has been implemented using Simio simulation software. It was created a dedicated library containing the following objects:

- **Equipment**: implementation of the generic equipment model presented above;
- **HPLCs**: liquid chromatography machines group;
- **GCs**: gas chromatography machines group;
- **KFs**: Karl Fischer titration machines group.

These objects have been used to build a complete model of the quality control laboratory system. As referred, when a sample is received in the laboratory it flows through the system towards the group of machines that can perform the test required. The scheduling is made within the equipment group and the sample is allocated to one of the available machines. To replicate the behavior of the real system and to evaluate its performances, every analysis has been scheduled to the same equipment that was used to run the test in the real laboratory within the considered time period. This choice allowed an easier validation of the developed model.

To check the correctness of the developed model and to visualize the scheduling, a 3D graphical model of the laboratory has been implemented and it is shown in figure 4.

Fig. 4. Quality Control laboratory model: Simio 3D view.

4. SIMULATION RESULTS

A simulation study aims to measure the performance of a real system and to evaluate how modifications on the system can affect its performance. In this study, the following aspects were considered:

- **System throughput**. Number of analyses performed over the total number of analyses required;
- **Equipment usage rate**. It is defined by the equipment operating time (EOT) over the total available time (TAT): $Usage_{\%} = \frac{EOT}{TAT}$;
- **System responsiveness**. Defined as the average time needed to process an analysis with the system in a steady state. Also referred to as *sample time in system*;
- **Tasks processing times**. Estimate of run time for each analysis task.

The discrete event simulation model was validated using a one year period data (referring to year 2015). Statistical independent replications of the simulation model, using the same input parameters, were run to retrieve information on the behavior of the model.

4.1 Model Validation and Performances

In order to verify the validity and accuracy of the developed simulation model, two metrics where used: *system throughput* and *equipment usage rate*.

Considering the system throughput defined above, the simulation model was able to perform all the analyses that where input to the system with no delay (see table 4). From this point of view, the model was effective to replicate the history. However, this metric is not conclusive since it only takes into account the total number of samples processed without considering the time needed to process them. To assess the validity of the developed simulation model, we considered the equipment usage rate that could be easily retrieved from the dedicated software of each equipment. Thus, the usage rate obtained with the simulation model has been compared with real data. In figure 5, it is shown a comparison of the usage rate of the most representative HPLC machines, between real and simulated data.

Table 4. System throughput.

Equip.	Number of analyses performed Real data	Model	Accuracy
HPLC	5968 (of 6099)	6079 (of 6099)	98.2%
GC	3899 (of 3910)	3910 (of 3910)	99.8%
PSA	3010 (of 3014)	3014 (of 3014)	99.9%
KF	1223 (of 1240)	1240 (of 1240)	98.7%
DSC	571 (of 581)	581 (of 581)	98.3%
XRPD	471 (of 479)	479 (of 479)	98.4%

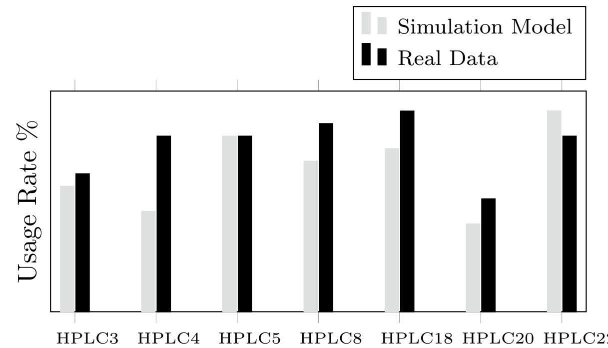

Fig. 5. HPLC usage rate(%) comparison between simulation and real data.

In average, the difference between real data and simulated data is within 1.5%. Similar values have been obtained considering other types of machines.

In a quality control laboratory, efficiency depends not only on the quality of the results but also on the time they are delivered. An important metric to evaluate the responsiveness of the system is the amount of time a sample spends in the laboratory before a result is generated. The average time in system for a sample is shown in table 5. This values take into account processing times and waiting times. The standard deviation is calculated among different simulation runs, and it does not represent variability in samples time in system. Given the lack of data, it is difficult to validate this result. Also, this result does not take into account the difference nature of a sample (i.e. in-process control, validation, stability, development). For instance, the average time in system for in-process control samples is 2 hours, while a sample with low priority (i.e. raw material) can be processed even after a week. This result may be representative considering a sample with medium priority and the system in a steady state.

Table 5. Average time in system for each sample type.

Sample Type	Average Time in System [h]
HPLC	31 ± 1
GC	10.0 ± 0.8
PSA	4.0 ± 0.5
KF	3.0 ± 0.5
DSC	5.0 ± 0.8
XRPD	6.0 ± 0.6

Processing times were input to the simulation model, and as a recall they were estimated considering deterministic and stochastic times. Table 6 shows average processing times obtained running the simulation model.

As expected, these results agree with the input distributions and with the analytical method reports. *System Preparation* and *Analysis Preparation* relate with the

Table 6. Simulation model average processing times. Time unite is minutes.

Equip.	System Prep.	System Suit.	Analysis Prep.	Analysis	Data Proc.
HPLC	207 ± 14	227 ± 11	96 ± 3	85 ± 1	59 ± 2
GC	67 ± 5	152 ± 5	48 ± 1	62 ± 1	39 ± 1
PSA	*n.a.*	*n.a.*	31 ± 1	32 ± 1	16 ± 1
KF	*n.a.*	*n.a.*	45 ± 3	30 ± 2	29 ± 2
DSC	*n.a.*	*n.a.*	47 ± 2	101 ± 3	24 ± 2
XRPD	*n.a.*	*n.a.*	46 ± 1	88 ± 3	30 ± 1

Sample Preparation entry in table 3. In particular, the time needed to prepare the system or the analysis is obtained multiplying the *Sample Preparation* time of table 3 and the number of samples/solutions needed to run the test.

4.2 Simulation with alternative scheduling policy

The simulation model allows to understand how modifications on the real system may affect its performances. With this purpose, it was tested a modified version of the model using a dedicated heuristic scheduling policy that takes into account two factors: *equipment loading* and *last analytical method used*. Preferably, a sample will be allocated to equipment that are already suitable to run the required analytical method. However, if an equipment suitable to run the right analytical method has already a high number of samples in queue (≥ 5), it will prefer a machine with a shorter queue.

With this configuration, two scenarios were tested. In the first scenario, the number of samples input to the system is the same of the considered time period. In scenario 2, the number of samples is increased of 50%. In table 7 it is shown the comparison between the two models (Real Model schedules analyses according to real data, while Modified Model uses the new scheduling policy).

Table 7. Average time in system for each sample type: comparison between models.

Sample Type	Average time in System [h] Scenario 1 Real M.	Scenario 1 Modified M.	Scenario 2 Real M.	Scenario 2 Modified M.
HPLC	31±1	15±1	57±1	69±8
GC	10.0±0.9	7.0±0.9	29±1	52±10
PSA	4±1	4.0±0.1	10±1	152±42
KF	3.0±0.2	3.0±0.2	3.7±1	18±2
DSC	5.0±0.2	7.0±0.2	10.0±0.7	237±42
XRPD	6±1	8.0±0.3	12±2	312±41

In the first scenario, the modified model is able to perform HPLC samples two times faster than the real model. Improvement in system responsiveness can be observed also on *Gas Chromatography samples*. For other classes of equipment the two models behave in a similar manner. The improvement observed is given by the fact that the new scheduling policy tries to take advantage of more machines avoiding long queues. In the second scenario, the modified model behaves worse than the real one. A reason for this is that there are not enough analysts in the system to process all the samples allocated to more machines. Many samples suffer from starvation and can wait a long time before an analyst is available to process them.

5. CONCLUSIONS

The quality control laboratory work flow can be described as a discrete event system that changes its state according to samples arrivals and timed operations defined on samples. Timed Petri net resulted in a powerful modeling tool that allowed an easy implementation of the QC laboratory model using a simulation software with 3D modeling capabilities. In order to make easy modifications to the model and to easily access to relevant information (i.e. processing times, arrival rates, etc.), input data was organized into a single database structure.

Simulations of the developed model showed that it is appropriate to mimic the real system and it can be used to estimate performance parameters and as a tool for planning, scheduling and decision making. Furthermore, simulation shows that modification of the scheduling policy may improve system performance and balance the loading of the machines. However, employees represent an important part of the work flow and scheduling effectiveness depends heavily upon the number of analysts present in the laboratory. On the other hand, in the simulated scenarios the number of equipment was appropriate to run all the analyses even considering a 50% of work overload.

The developed simulation model can be applied to a wide range of laboratory organizations and classes of instruments. It can be used to simulate different scenarios and as a support tool for decision making (i.e. planning and scheduling). The proposed case study highlighted the importance of the estimation of the correct number of resources (machines and equipment) and the effectiveness of dedicated heuristics for analyses scheduling. Suggestions to improve the efficiency of the simulated lab are the development of dedicated algorithms for planning and scheduling that rely on the database structure proposed in this work.

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